Judging Financial Fair Play Fairly: The Economics of Burnley vs Everton

Thomas Peeters, James Reade, Carl Singleton and Stefan Szymanski

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Introduction

- Part I (Stefan): the Burnley v Everton case - background, economic damages and the loss of a chance
- Part II (James): estimating the probability that Burnley would have been relegated in the 2021/22 season

Background: Everton's breach of PSR

- Profit and Sustainability Rules (PSR) adopted by English Premier League (EPL) in 2015/16.
- Permitted allowable losses of £105 million over three seasons if covered by secure funding
- Over the three season period 2019/20 to 2021/22 Everton was found to have breached the limit by £19.5 million
- EPL Commission punished Everton with a 6 point penalty (reduced from 10 on appeal) deducted in 2023/24 season.

Background: Burnley's Rule W Claim

- In 2021/22 Burnley was relegated from the EPL in 18th place with 35 points; Everton finished in 16th place on 39 points
- Rule W permits clubs to claim compensation for damages for breaches of PSR rules
- Burnley claimed they were harmed by Everton's breach of the PSR rules
- Burnley claimed that in the counterfactual case where Everton remained within the spending limits, Everton would have been relegated, not Burnley

Background: Compensation claims for breach of PSR rules

- Burnley sued in 2024; following hearings in 2025, the Commission published its decision in June 2026
- There have been six other clubs found to have breached PSR rules (Birmingham City, Nottingham Forest, Derby County, West Bromwich Albion, Leicester City and most significantly, **Manchester City**)
- Eight clubs have been punished for breaches of the English Football League version of PSR (Queen's Park Rangers, AFC Bournemouth, Leicester City, Nottingham Forest, Leeds United, Blackburn Rovers, Fulham, Cardiff City)
- Manchester United, Arsenal and Liverpool have indicated they may sue Manchester City
- PSR rules replaced by Squad Cost Rule in 2026/27, but compensation rules remain.

Background: Damages under English law

- Under common law, a plaintiff can recover financial damages for breach of contract
- The logic is almost identical to cases of negligence (e.g. medical)
  - The plaintiff must establish "causation" – meaning that in the counterfactual world where the breach/negligence did not occur, there would have been no harm
  - If causation is established, the court decides the "quantum" – the amount of the damages to be paid

Background: The standard for causation in English law

- The plaintiff must establish that under the counterfactual it was "more likely than not" that the harm would not have occurred:

$$D = \begin{cases} \boldsymbol{V_{no\ harm} - V_{harm}}, & \text{if } \boldsymbol{P^C_{no\ harm} > 0.5}, \\ \boldsymbol{0}, & \textbf{otherwise}, \end{cases} \tag{1}$$

- Here:
  - $[V_{no\ harm} - V_{harm}]$ is the quantum,
  - $P^C_{no\ harm}$ is the counterfactual probability of no harm.

- This is at odds with the usual economic (expected value) formulation

$$\boldsymbol{D^* = \left[P^A_{no\ harm} - P^C_{no\ harm}\right] \times [V_{no\ harm} - V_{harm}]}, \tag{2}$$

- Here, $P^A_{no\ harm}$ is the probability of no harm in actuality.

Background: The English standard is problematic

- Gregg v Scott, 2005
- Gregg was misdiagnosed by his doctor (Scott) which caused a delay in the discovery of his cancer by one year
- It was agreed that the misdiagnosis caused the ten-year survival probability of Gregg to fall from 42% to 18%.
- Because Gregg's survival was not "more likely than not" under the counterfactual ($P^{C}_{no\ harm} < 0.5$), his case failed and no damages were paid

Background: Judicial justification

1. A high bar limits frivolous claims
2. Inherent uncertainty and ignorance – in medical cases the probabilities are not inherent, merely a reflection of our lack of knowledge

   But highly controversial – see e.g. (Stapleton, 1988, 2005; Peel, 2003; Miller, 2006)

Background: A better approach – the loss of a chance

- Chaplin v Hicks, 1911
- Hicks offered to enter women in a beauty contest if they sent him a picture and he deemed it pretty enough
- Chaplin sent a picture and Hicks agreed to enter her, but he forgot
- Chaplin sued for the loss of chance to compete
- Uncertainty was inherent – she lost a chance to win
- She won her case in court and was paid damages – the winning prize adjusted for the probability of winning

Background: Application to football

- The uncertainty of outcome hypothesis is the cornerstone of sports economics (uncertainty is inherent)
- The harm to Burnley caused by Everton was not relegation, but the increased probability of relegation

$$D^* = \left[P(B_R^A) - P(B_R^C)\right] \times [V_{NR} - V_R] \ , \qquad (3)$$

- $P(B_R^A)$ and $P(B_R^C)$ are Burnley's probability of being relegated given Everton's actual spending and the counterfactual.
- Loss of chance is the right principle to apply in damages cases such as Burnley v Everton

Background: The (erroneous) Burnley decision

- The Commission decided that under the counterfactual it was more likely than not that Everton would have been relegated – the range presented by Burnley's experts was between 51% and 70%
- As a result Everton were liable for the difference in value between Burnley remaining in the EPL and being relegated to the EPL (quantum)
- Quantum was calculated as EBITDA + net player trading
- The Commission awarded a quantum of £35 million

Background: Why $P(B_R^A) \neq 1$

- The Commission's approach attributes the entirety of the actual outcome (Burnley's relegation) to Everton's overspend
- In reality, the actions of all clubs in the EPL contributed to the outcome
- English courts in practice often apply the loss of chance logic when outcomes depend on the actions of third parties as well

Background: the correct framing - Burnley's loss of a chance

- Everyone knew at the beginning of the 2021/22 that Burnley was among the most likely to be relegated:
  - BetMGM 22.9% for relegation.
  - PointsBet (via NBC): ≈29%.
  - Oddschecker via The Week and The Mirror: ≈29%
- We estimate the following: by how much did Burnley's relegation probability increase assuming that all of Everton's excess spending applied to the season 2021/22?
- (we do not consider quantum)

Methodology

- Model to estimate cumulative probability distribution:
  - Team i finishing at least as 'high' as position r.

$$F_i(r) = \sum_{k=1}^{r} p_{ik} \ , \tag{4}$$

  - p_{ik} is probability team i finishes in position k.
  - Note $r = 20, \ldots, 1$.
- $F_i(r)$ is probability team i finishes in position r or better.
  - $F_i(17)$: probability of avoiding relegation,
  - $F_i(1) = p_{i1}$: probability of winning title.

Methodology

- Define:
  - X_i: Total league points for team i.
  - G_i: Goals scored by team i.
  - C_i: Goals concended by team i.
  - $D_i = G_i - C_i$: Goal difference.
- League standing probabilities can be rewritten as:

$$F_i\left(r \mid W_i, \mathbf{W}_{-i}\right) = P\left(\sum_{j \neq i}\left[1_{\{X_j > X_i\}} + 1_{\{X_j = X_i,\, D_j > D_i\}}\right] \leq r - 1\right). \tag{5}$$

- W_i: team i wage bill.
- $\mathbf{W}_{-i}$ wage bills of all other teams.

Methodology

- W_i and $\mathbf{W}_{-i}$ reflects central sports economics concept:
  - Team success depends also on resource of competitors.
  - English leagues (Szymanski and Smith, 1997; Hall et al., 2002; Carmichael et al., 2011).
  - European leagues (Forrest and Simmons, 2002; Franck and Nüesch, 2011; Peeters and Szymanski, 2014).
- Given wage bill produces higher position if rivals spend less.
- Wage bills provide basis for counterfactual analysis:
  1. Robust relationship between wages and outcomes.
  2. Change team wage bill holding all other factors fixed.

Methodology

- Points accumulated from wins and draws.
- Teams in EPL season play each other twice:

$$X_i = \sum_{j \neq i} \pi\left(Y_{ij}\right) + \sum_{j \neq i} \tilde{\pi}\left(Y_{ji}\right) \ ,$$

  - Y_{ij} (Y_{ji}): Match outcome between home (away) team i and away (home) team j.
  - $\pi\left(\cdot\right)$, $\tilde{\pi}\left(\cdot\right)$ translateinto points.
- Express probabilities of outcomes as:

$$P\left(Y_{ij} = y \mid \boldsymbol{W}, \boldsymbol{Z}\right) = p_y\left(\ W_i/W_j \ , \ Z_{i1}/Z_{j1} \ , \ldots , \ Z_{ik}/Z_{jk} \ \right) \ . \tag{6}$$

Methodology

- Model Y_{ij} via goal arrival (Maher, 1982; Dixon and Coles, 1997):
  - Poisson regression model of home (h_{ij}), away (a_{ij}) goals.
  - Use to calculate probabilities of outcomes.
- Karlis and Ntzoufras (2003): bivariate Poisson model.
- Class of model studied extensively (e.g., Forrest et al., 2005; Koopman and Lit, 2015; Reade et al., 2021.

Methodology

- Assume home (h_{ij}, away (a_{ij}) goals Poisson distributed: Data

$$P(H_{ij} = a_{ij}) = \frac{\lambda_{ij}^{h} e^{-\lambda_{ij}^{h}}}{h_{ij}!}, \tag{7}$$

$$P(A_{ij} = a_{ij}) = \frac{\lambda_{ij}^{a} e^{-\lambda_{ij}^{a}}}{a_{ij}!}. \tag{8}$$

- Expected goals depends on own and opposing team strengths:

$$\log(\lambda_{ij}^{h}) = \alpha_i + \beta_j + \gamma, \tag{9}$$

$$\log(\lambda_{ij}^{a}) = \alpha_j + \beta_i. \tag{10}$$

- αs are attacking strengths, βs defensive. γ home advantage.

Methodology

- With estimates for (11) and (12) can generate $\widehat{\lambda}_{ij}^{h}$ and $\widehat{\lambda}_{ij}^{a}$.
- Can use $\widehat{\lambda}_{ij}^{h}$ and $\widehat{\lambda}_{ij}^{a}$ to generate random numbers:
  - Consistent with expected attacking/defensive strengths.
  - Simulations of football matches.
  - With thousands of replications can generate distributions.
- Simulations allow frequentist estimates for $P(Y_{ij} = y)$, $F(r)$.

Methodology

- Simulations must rely on **robust** model.
  - We first establish model quality.
  - Run simulations from 2001/02 to 2024/25.
  - Compare to actual outcomes.
- Equations (11) and (12) need W_i, W_j.
  - Bradley-Terry formation a common strategy.
  - But does not allow different team efficiencies.
  - To isolate wage spend effect we omit fixed effects.
- Our model:

$$\log(\lambda_{ij}^{h}) = \alpha_i \log(W_i) + \beta_j \log(W_j) + \gamma, \tag{11}$$

$$\log(\lambda_{ij}^{a}) = \alpha_j \log(W_j) + \beta_i \log(W_i). \tag{12}$$

- Counterfactual: Reduce Everton W_i by £19.5m.

Results: Robustness

- Match-level predictions: Results
- Season-level outcomes: Points Position Position Box

Results: The 2021/2022 Season

- Points Results Plot of changes
- Positions Results

	Everton		Burnley	
	Actual	Counterfactual	Actual	Counterfactual
Points (mean)	49.6	47.3	41.5	41.6
Points (st dev)	7.6	7.5	7.4	7.4
Position (mean)	11.9	12.9	14.6	14.5
Position (st dev)	3.3	3.4	3.3	3.3

Results: The 2021/2022 Season

- Relegation probabilities Table Plot of changes

	$1 - F(17)$				
	Actual	Counter-factual	Difference (causation)	Quantum	Damages
Commission	1	0	1	£35m	£35m
Model	0.219	0.214	0.005	£35m	£0.175m
Difference	0.781	-0.214	0.995	0	£34.825m

Conclusions

- We review basis of award in Burnley v Everton.
- We make two arguments:
  1. As a matter of law, the methodology adopted was inappropriate and inconsistent with established methodologies for the evaluation of losses in cases where chance and uncertainty of outcome are inherent in the calculation.
  2. The specific methodology presented by the economic experts for Burnley and accepted by the Commission were flawed from the perspective of statistical analysis.

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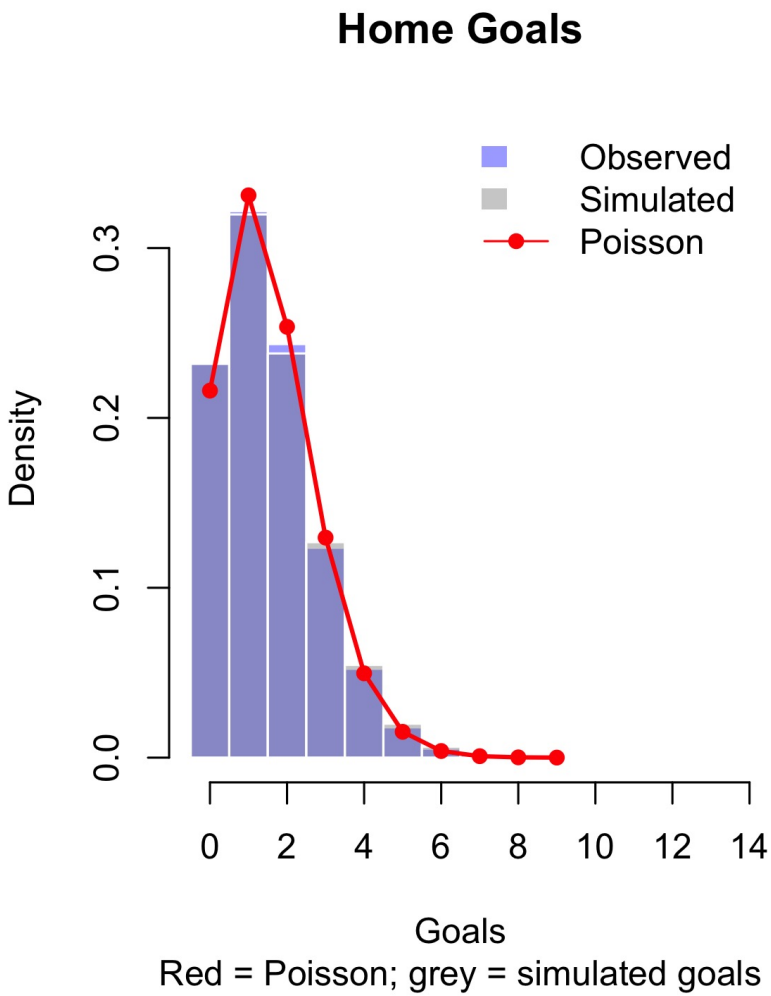
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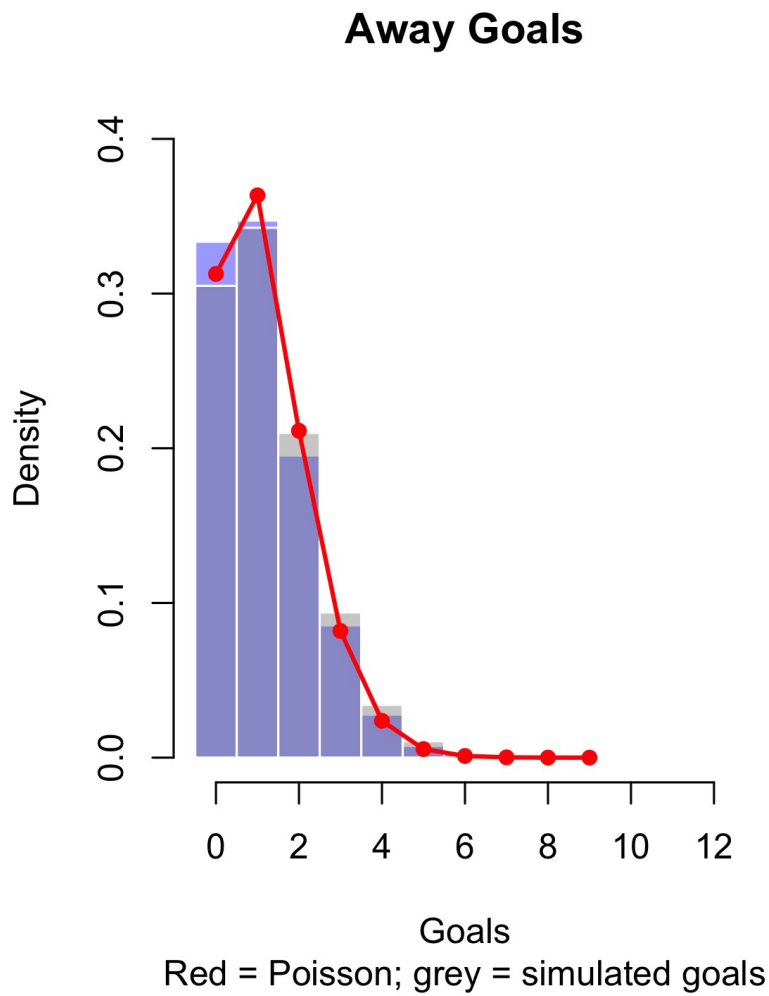
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Poisson

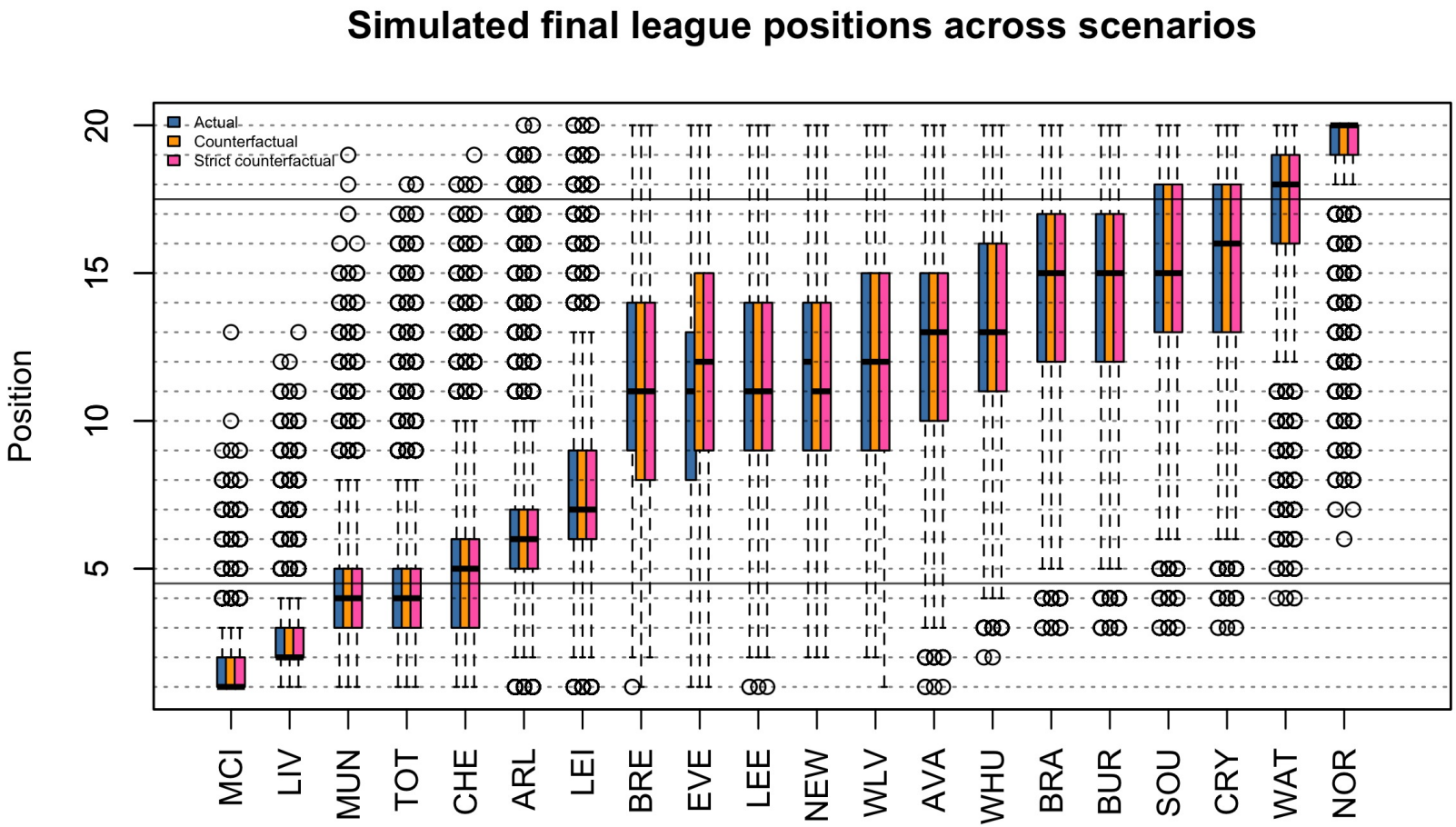

Home Goals
Observed
Simulated
Poisson
Density
0.0
0.1
0.2
0.3
0
2
4
6
8
10
12
14
Goals
Red = Poisson; grey = simulated goals


Away Goals
Density
0.0
0.1
0.2
0.3
0.4
0
2
4
6
8
10
12
Goals
Red = Poisson; grey = simulated goals

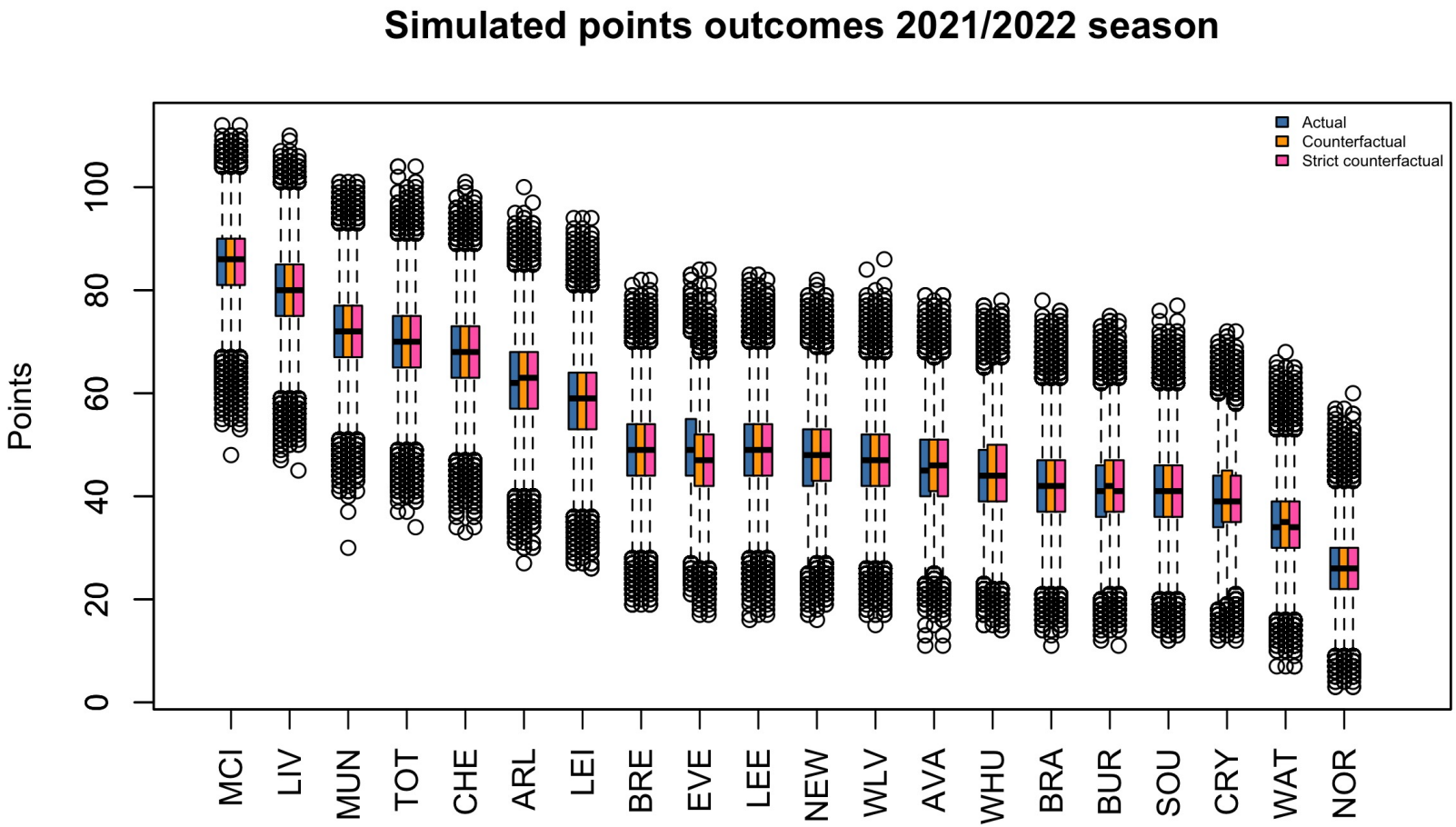
Relegation Probabilities

team	actual	counterfactual	difference
Norwich City	93.86	93.72	-0.14
Watford	59.66	59.16	-0.50
Crystal Palace	31.57	30.99	-0.58
Southampton	25.95	25.68	-0.28
Burnley	21.89	21.39	-0.50
Brighton and Hove Albion	19.24	19.05	-0.20
West Ham United	12.55	12.39	-0.16
Aston Villa	9.92	9.78	-0.14
Wolverhampton Wanderers	6.92	6.75	-0.16
Newcastle United	5.81	5.75	-0.07
Brentford	4.54	4.38	-0.16
Leeds United	4.45	4.35	-0.10
Everton	3.39	6.36	2.97
Leicester City	0.20	0.21	0.01
Arsenal	0.04	0.04	0.00
Chelsea	0.00	0.01	0.00
Liverpool	0.00	0.00	0.00
Manchester City	0.00	0.00	0.00
Manchester United	0.00	0.00	0.00
Tottenham Hotspur	0.00	0.00	0.00

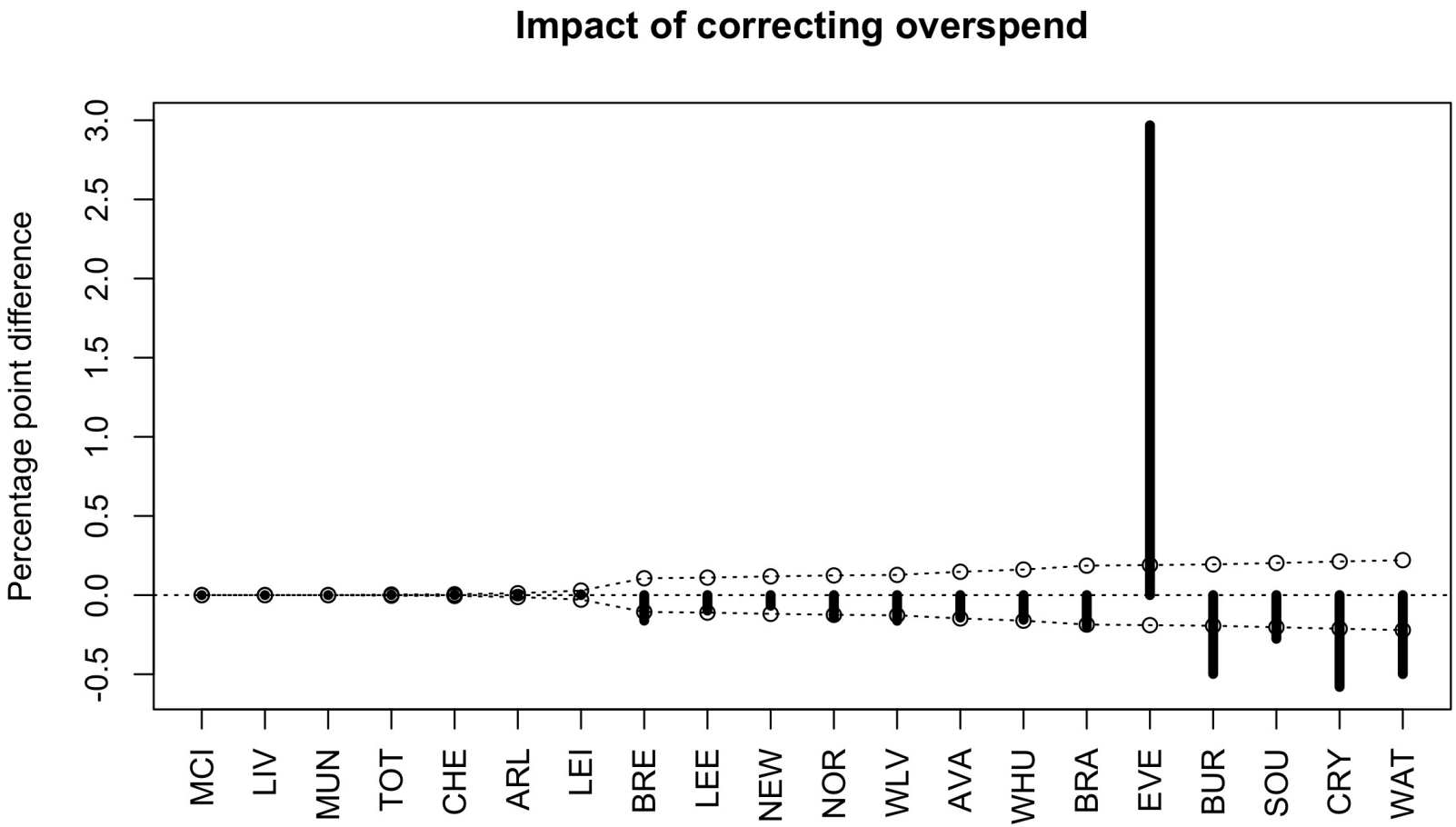
2021/22: Positions



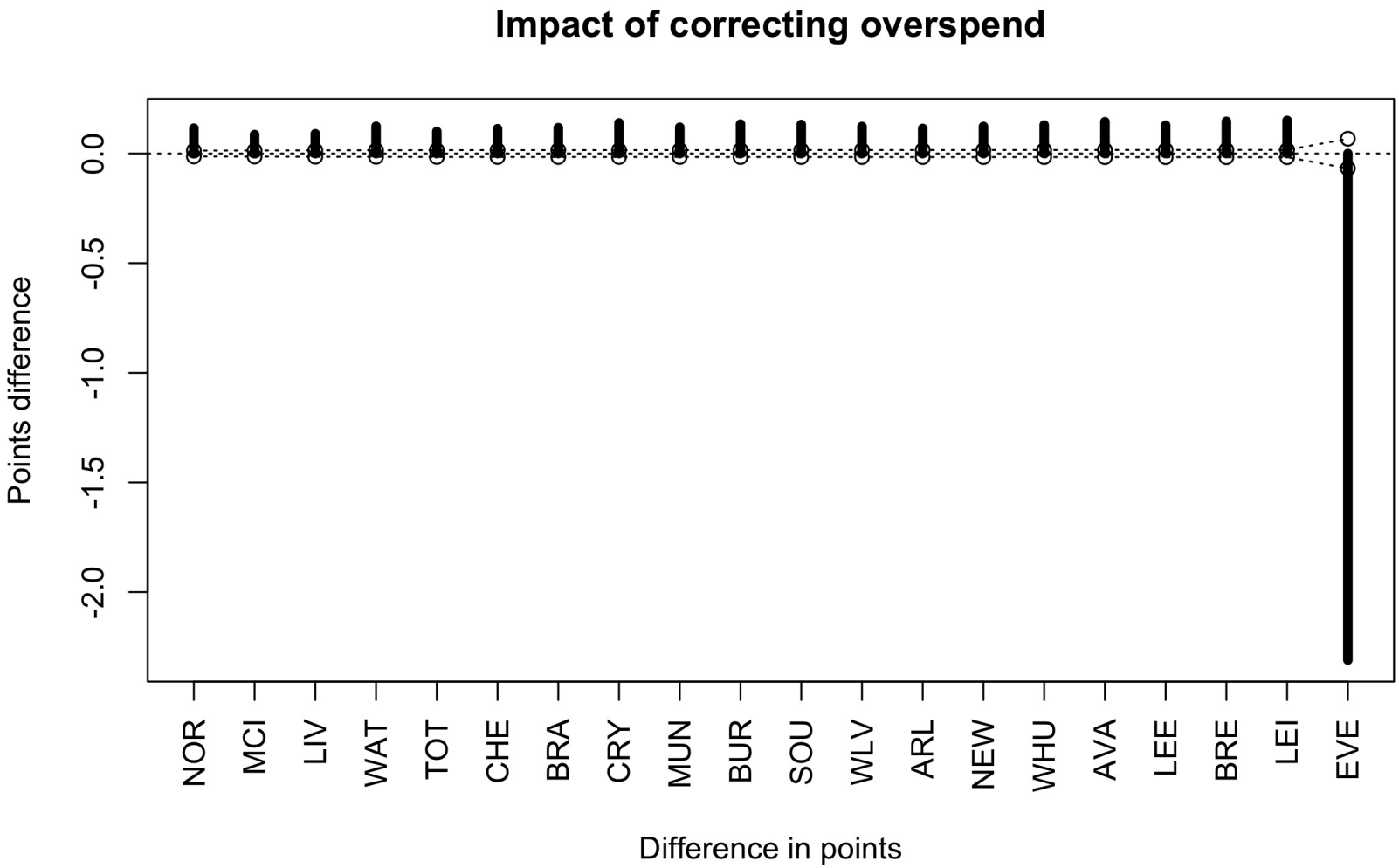
2021/22: Points



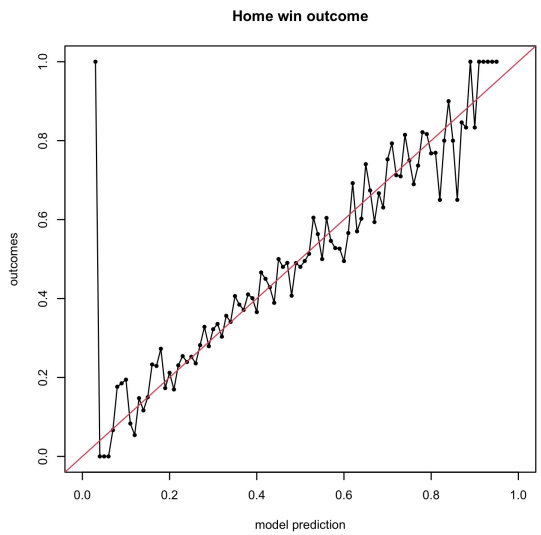
2021/22: Probability of relegation

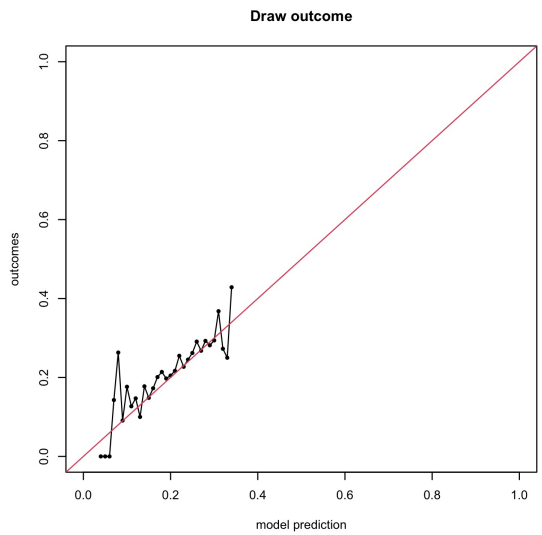


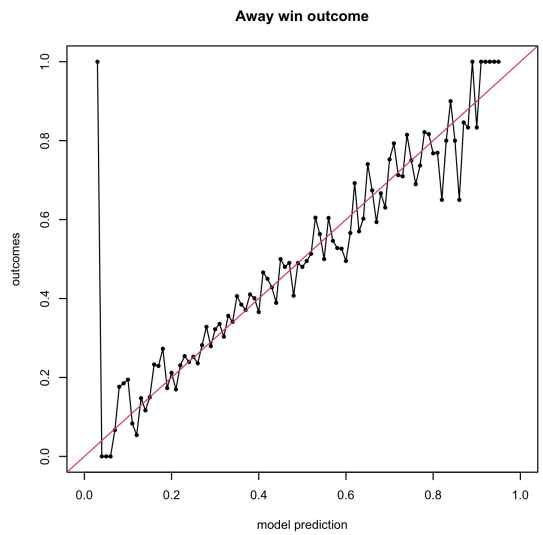
2021/22: Points difference



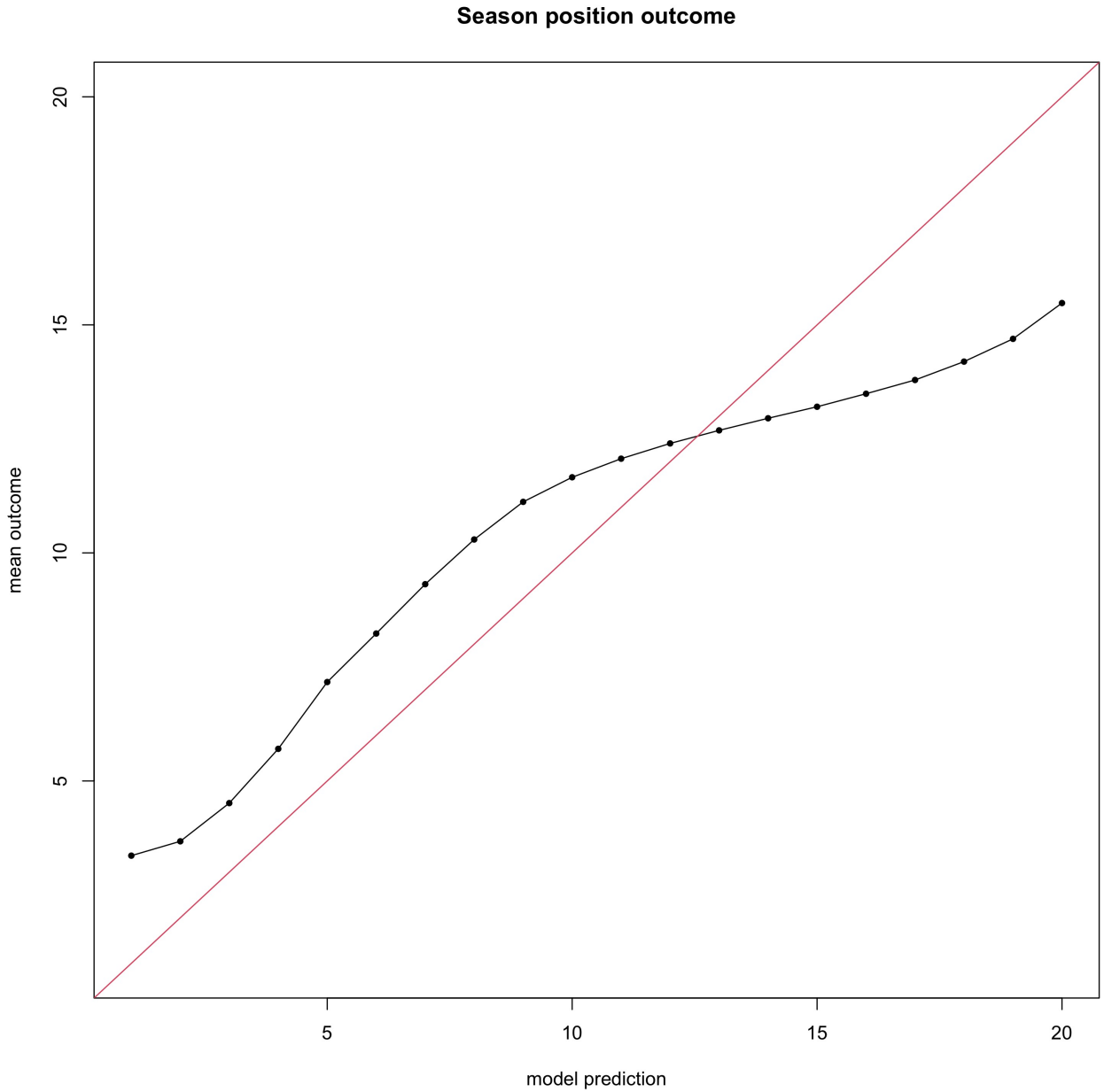
Matches: Home win, draw and away win outcomes



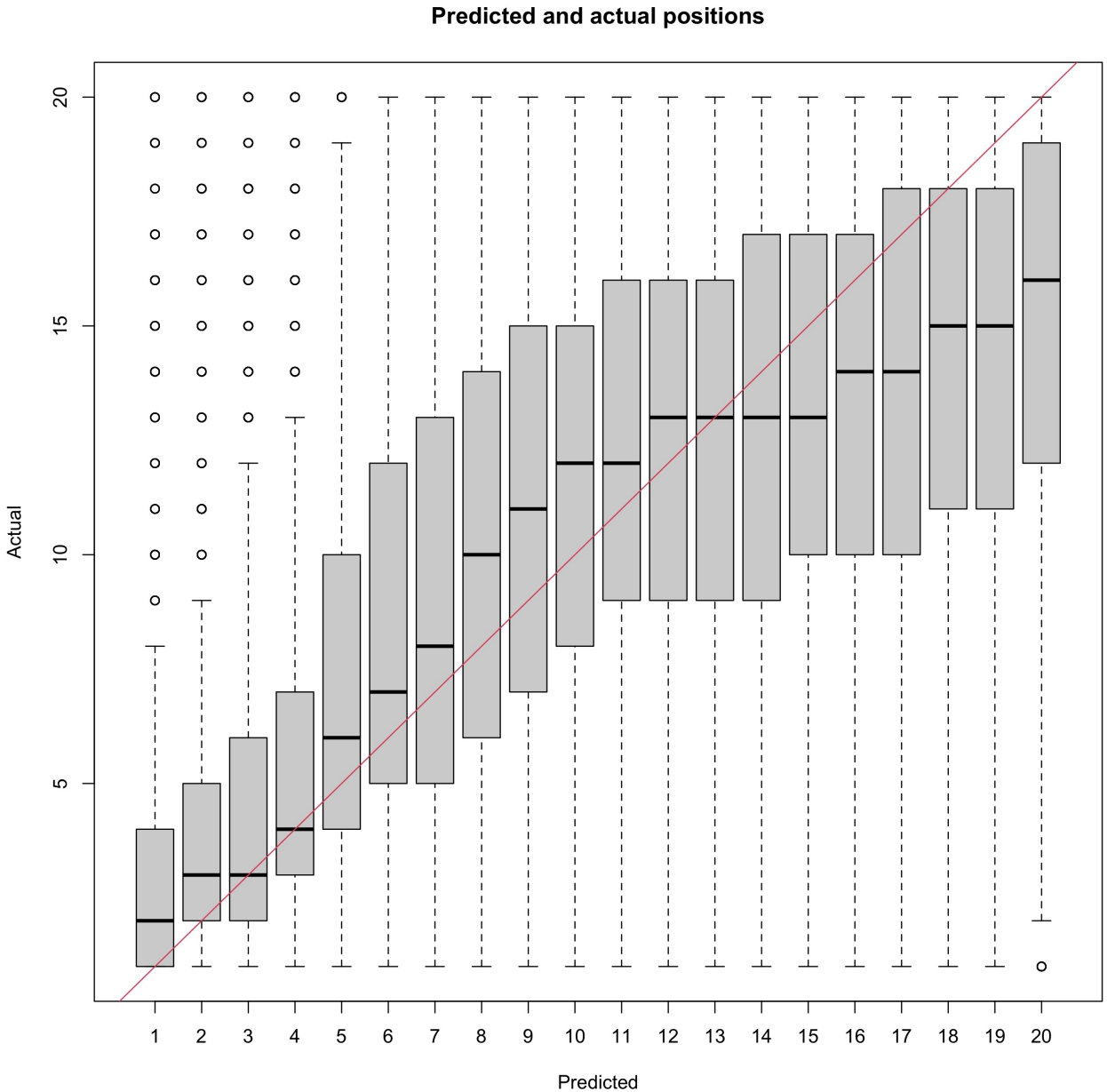




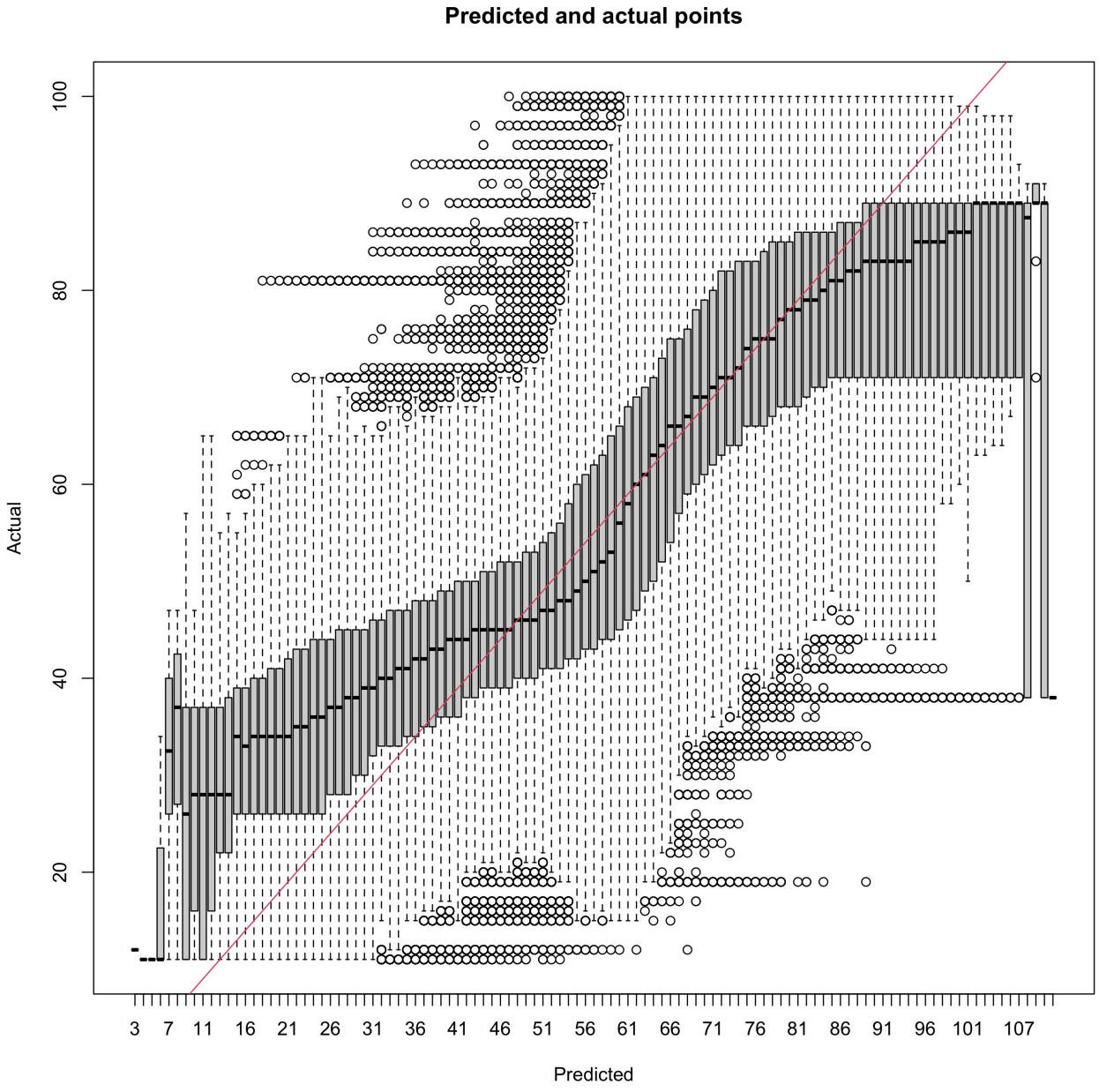
Season: Position



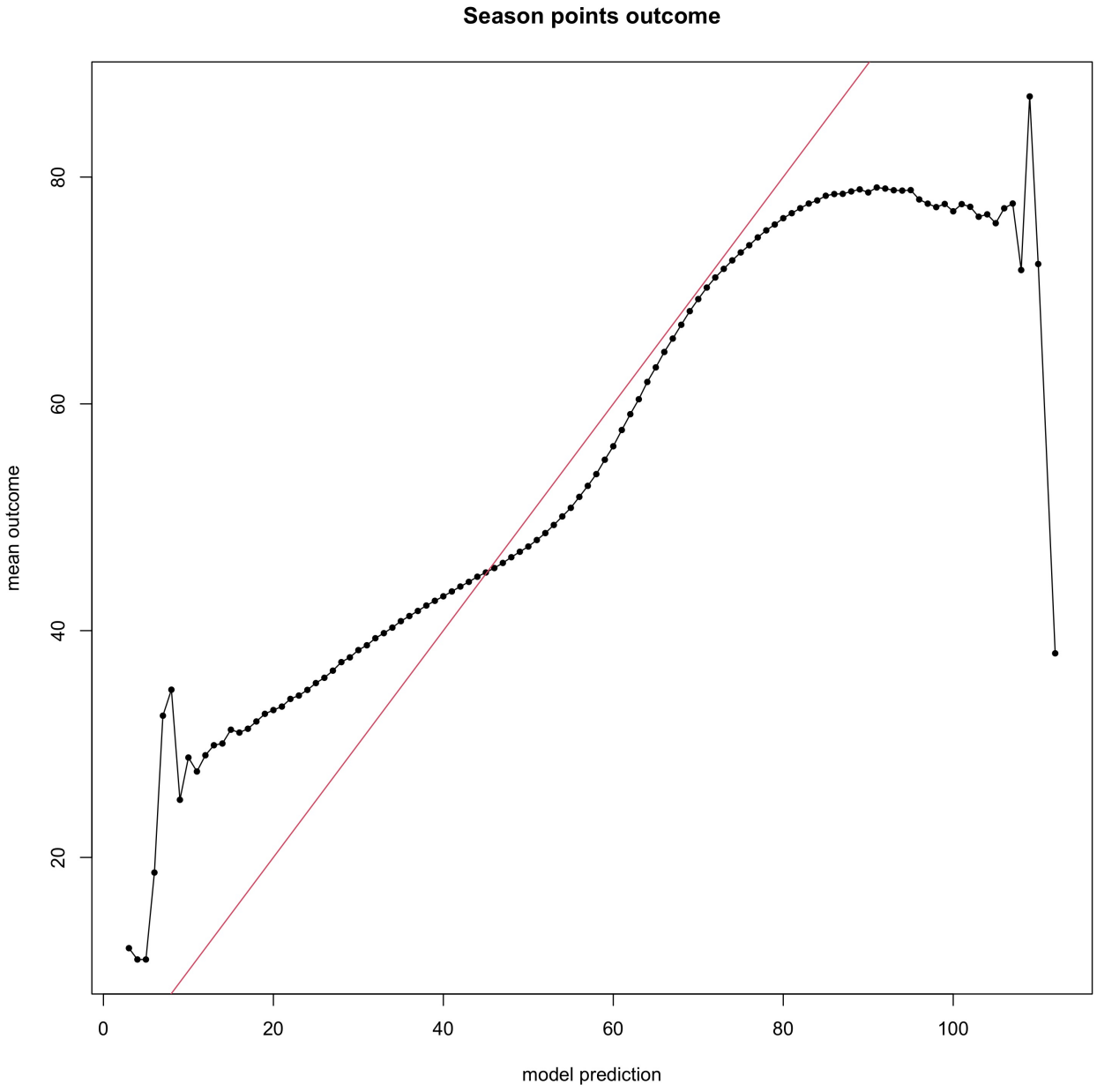
Season: Position



Season: Points



Poisson


Season points outcome
mean outcome
model prediction
0
20
40
60
80
100
20
40
60
80